

Final Comprehensive Examination – Medicinal Plants

Lectures 5–12 (Blueprint-Based Version)

Time: 3 Hours

Total Marks: 100

Marking Scheme:

- 50 Questions × 2 Marks = 100 Marks

Instructions: - Answer all questions. - Choose the single best answer.

- Coverage: Lectures 5–12 only.

Section A: Leaf Morphology & Microscopy (Lecture 5) – 6 Questions

Q1.

The leaf is best defined as:

- A. A reproductive organ B. A lateral outgrowth from the stem
- C. A modified root D. An underground stem

Q2.

Anomocytic stomata are characterized by:

- A. Presence of two subsidiary cells B. Subsidiary cells parallel to guard cells
- C. Absence of distinct subsidiary cells D. Unequal subsidiary cells

Q3.

Which tissue is mainly responsible for photosynthesis in leaves?

- A. Spongy mesophyll B. Palisade mesophyll C. Epidermis D. Xylem

Q4.

Beaded anticlinal walls are best observed using:

- A. Organoleptic evaluation B. Macroscopy C. Microscopy D. Chemical testing

Q5.

The main function of cuticle is to:

A. Increase transpiration B. Prevent water loss C. Store food D. Transport minerals

Q6.

Which feature is MOST reliable for microscopic identification of powdered leaves?

A. Color B. Taste C. Stomatal type D. Leaf size

Section B: Senna & GIT Drugs (Lecture 6) – 6 Questions

Q7.

The laxative action of Senna is due to:

A. Alkaloids B. Flavonoids C. Anthraquinone glycosides D. Saponins

Q8.

Sennosides are considered prodrugs because they:

A. Act directly in stomach B. Are activated by intestinal bacteria

C. Are rapidly absorbed D. Are excreted unchanged

Q9.

Senna is contraindicated in intestinal obstruction because it:

A. Causes sedation B. Increases peristalsis and water secretion

C. Reduces motility D. Causes gastric erosion

Q10.

Chronic use of Senna may lead to:

A. Hyperkalemia B. Hypokalemia C. Hypercalcemia D. Hyponatremia

Q11.

The major site of Senna action is the:

A. Stomach B. Small intestine C. Colon D. Liver

Q12.

Which adverse effect is commonly associated with Senna?

A. Dry mouth B. Gripping abdominal pain C. Bronchoconstriction D. Sedation

Section C: ANS & Anticholinergic Plants (Lecture 7) – 7 Questions

Q13.

Belladonna and Datura exert their main effect by:

A. Stimulating muscarinic receptors B. Blocking muscarinic receptors
C. Stimulating nicotinic receptors D. Inhibiting acetylcholine synthesis

Q14.

A classic feature of anticholinergic toxicity is:

A. Diarrhea B. Miosis C. Dry mouth and blurred vision D. Bradycardia

Q15.

Hyoscine is preferred in motion sickness because it:

A. Has strong cardiac action B. Produces CNS sedation
C. Is free from side effects D. Has shorter half-life only

Q16.

Which plant is commonly known as Deadly nightshade?

A. Datura stramonium B. Atropa belladonna
C. Hyoscyamus muticus D. Pilocarpus jaborandi

Q17.

Pilocarpus differs from Belladonna by being:

A. Anticholinergic B. Cholinergic C. Cardiotonic D. CNS depressant

Q18.

Which condition contraindicates anticholinergic drugs?

A. Motion sickness B. Peptic ulcer C. Glaucoma D. Hypotension

Q19.

The pharmacological action linking Belladonna, Datura, and Hyoscyamus is:

A. Cardiotonic B. Anticholinergic C. Expectorant D. Diuretic



Section D: Tropane Alkaloids & Tea (Lecture 8) – 7 Questions

Q20.

The Vitali–Morin reaction is specific for:

A. Cardiac glycosides B. Tropane alkaloids C. Anthraquinones D. Saponins

Q21.

Hyoscyamus alkaloids mainly produce:

A. Cholinergic effects B. Anticholinergic effects

C. Adrenergic effects D. Cardiotonic effects

Q22.

Atropine formation from hyoscine occurs by:

A. Enzymatic hydrolysis B. Isomerization C. Oxidation D. Reduction

Q23.

Tea leaves are obtained mainly from:

A. Roots B. Mature lower leaves C. Young leaves and buds D. Stems

Q24.

The stimulant effect of tea is due to:

A. Tannins B. Flavonoids C. Caffeine D. Saponins

Q25.

Which factor decreases tea quality?

A. Young leaves B. Rapid drying C. Leaf maturity D. Proper fermentation

Q26.

Hyoscine is therapeutically preferred over atropine because it:

- A. Has stronger cardiac action
 - B. Produces more CNS sedation
 - C. Is non-toxic
 - D. Acts as laxative
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Section E: Uva Ursi & Urinary Tract (Lecture 9) – 6 Questions

Q27.

The main active constituent of Uva Ursi is:

- A. Tannins
- B. Arbutin
- C. Flavonoids
- D. Saponins

Q28.

Hydroquinone released from arbutin acts mainly as:

- A. Diuretic
- B. Antibacterial
- C. Antispasmodic
- D. Sedative

Q29.

Uva Ursi should not be used long-term due to:

- A. Renal stones
- B. High tannin content
- C. Hypokalemia
- D. Cardiac toxicity

Q30.

The antibacterial action of Uva Ursi is optimal when urine is:

- A. Acidic
- B. Neutral
- C. Alkaline
- D. Concentrated

Q31.

Greenish-brown urine discoloration during therapy indicates:

- A. Toxic reaction
- B. Infection
- C. Oxidation of hydroquinone
- D. Liver damage

Q32.

Uva Ursi belongs to which family?

- A. Myrtaceae
 - B. Solanaceae
 - C. Ericaceae
 - D. Fabaceae
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Section F: Respiratory Plants (Lecture 10) – 6 Questions

Q33.

Which plant contains saponins acting as expectorants?

A. Guava B. Ivy C. Eucalyptus D. Tea

Q34.

Eucalyptus oil relieves nasal congestion mainly by:

A. CNS depression B. Acting on cold receptors
C. Increasing mucus viscosity D. Causing sedation

Q35.

Guava leaves are mainly used for their:

A. Laxative effect B. Astringent effect C. Cardiotonic effect D. Sedative effect

Q36.

Which plant is most suitable for productive cough?

A. Guava B. Ivy C. Tea D. Senna

Q37.

Overuse of eucalyptus oil may cause:

A. Hypotension B. Respiratory irritation C. Sedation D. Diuresis

Q38.

The expectorant action of ivy is due to:

A. CNS stimulation B. Reflex gastric irritation
C. Anticholinergic action D. Direct cardiotonic effect

♡ Section G: Cardiac Glycosides – Digitalis (Lecture 11) – 8 Questions

Q39.

Digitalis glycosides exert their main effect by:

- A. Decreasing heart rate only
- B. Increasing myocardial contractility
- C. Causing vasodilation
- D. Producing diuresis only

Q40.

The aglycone part of Digitalis glycosides is:

- A. Flavonoid nucleus
- B. Alkaloid nucleus
- C. Steroidal nucleus with lactone ring
- D. Purine nucleus

Q41.

Rapid drying of Digitalis leaves is essential to:

- A. Improve color
- B. Prevent enzymatic hydrolysis
- C. Increase solubility
- D. Enhance absorption

Q42.

Which Digitalis species contains acetylated glycosides?

- A. Digitalis purpurea
- B. Digitalis lanata
- C. Digitalis lutea
- D. Digitalis ferruginea

Q43.

The sugar part of cardiac glycosides mainly affects:

- A. Pharmacological activity
- B. Solubility and pharmacokinetics
- C. Toxicity only
- D. Enzyme inhibition

Q44.

Digitalis toxicity is mainly due to:

- A. Rapid elimination
- B. Accumulation in tissues
- C. Poor absorption
- D. Low potency

Q45.

Which symptom is associated with Digitalis toxicity?

- A. Diarrhea
- B. Visual disturbances
- C. Bronchospasm
- D. Hypoglycemia

Q46.

Digitalis requires careful dose monitoring because it:

- A. Has wide therapeutic index B. Has narrow therapeutic index
 - C. Is rapidly excreted D. Is weakly active
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Section H: Plant Taxonomy (Lecture 12) – 4 Questions

Q47.

Plant taxonomy deals mainly with:

- A. Physiology B. Identification and classification C. Pharmacology D. Cultivation

Q48.

Cryptogams reproduce mainly by:

- A. Seeds B. Flowers C. Spores D. Fruits

Q49.

Monocotyledons are characterized by:

- A. Two cotyledons B. Reticulate venation C. Parallel venation D. Secondary growth

Q50.

The lowest taxonomic rank is:

- A. Family B. Genus C. Species D. Order
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End of Exam

Model Answers & Explanations

Final Comprehensive Exam – Medicinal Plants (Lectures 5–12)

Purpose: Instructor use, grading reference, and final student revision

Based strictly on the Blueprint-Based Final Exam

Section-wise Model Answers & Detailed Explanations

Q1. B

A leaf is defined botanically as a lateral outgrowth from the stem with a definite structure and function, mainly photosynthesis. Other options describe roots or stems.

Q2. C

Anomocytic stomata lack distinct subsidiary cells and are surrounded by ordinary epidermal cells.

Q3. B

Palisade mesophyll contains elongated cells rich in chloroplasts, making it the main photosynthetic tissue.

Q4. C

Beaded anticlinal walls are microscopic characters visible only under the microscope, not by naked eye or chemical tests.

Q5. B

The cuticle is a waxy, impermeable layer that minimizes water loss and protects the leaf.

Q6. C

Stomatal type is a stable microscopic feature and is highly diagnostic for powdered leaf identification.

Q7. C

Senna contains anthraquinone glycosides (sennosides) responsible for its laxative effect.

Q8. B

Sennosides are inactive until metabolized by colonic bacteria into active anthraquinone aglycones.

Q9. B

Senna increases both intestinal peristalsis and water secretion, which is dangerous in intestinal obstruction.

Q10. B

Chronic Senna use leads to potassium loss, resulting in hypokalemia.

Q11. C

Senna acts mainly in the colon after bacterial activation of sennosides.

Q12. B

Gripping abdominal pain is a common adverse effect due to increased peristalsis.

Q13. B

Belladonna and Datura block muscarinic receptors, producing anticholinergic effects.

Q14. C

Dry mouth and blurred vision are hallmark signs of anticholinergic toxicity.

Q15. B

Hyoscine causes greater CNS depression, making it useful in motion sickness.

Q16. B

Atropa belladonna is commonly known as Deadly nightshade.

Q17. B

Pilocarpus is cholinergic, unlike Belladonna which is anticholinergic.

Q18. C

Anticholinergics increase intraocular pressure, worsening glaucoma.

Q19. B

Belladonna, Datura, and Hyoscyamus all share anticholinergic action.

Q20. B

Vitali–Morin reaction is a specific chemical test for tropane alkaloids.

Q21. B

Hyoscyamus alkaloids block muscarinic receptors producing anticholinergic effects.

Q22. B

Atropine can be formed from hyoscyne by isomerization under harsh conditions.

Q23. C

Tea is prepared from young leaves and terminal buds for best quality.

Q24. C

Caffeine is responsible for the CNS stimulant effect of tea.

Q25. C

Older leaves reduce tea quality due to increased fiber and reduced active constituents.

Q26. B

Hyoscyne is preferred due to stronger CNS sedative effect.

Q27. B

Arbutin is the major active constituent of Uva Ursi.

Q28. B

Hydroquinone released from arbutin exerts antibacterial action in the urinary tract.

Q29. B

High tannin content causes gastric irritation on prolonged use.

Q30. C

Hydroquinone acts optimally in alkaline urine.

Q31. C

Oxidation of hydroquinone causes harmless greenish-brown urine discoloration.

Q32. C

Uva Ursi belongs to the Ericaceae family.

Q33. B

Ivy contains saponins that act as expectorants.

Q34. B

Eucalyptus oil stimulates cold receptors producing decongestant effect.

Q35. B

Guava leaves are rich in tannins, giving astringent action.

Q36. B

Ivy is most suitable for productive cough due to mucolytic effect.

Q37. B

Excess eucalyptus oil may irritate respiratory mucosa.

Q38. B

Saponins act reflexly by irritating gastric mucosa.

Q39. B

Digitalis increases myocardial contractility (positive inotropic effect).

Q40. C

Cardiac glycosides possess a steroidal nucleus with a lactone ring.

Q41. B

Rapid drying prevents enzymatic degradation of glycosides.

Q42. B

Digitalis lanata contains acetylated glycosides.

Q43. B

Sugar moiety affects solubility and pharmacokinetics of cardiac glycosides.

Q44. B

Digitalis accumulates in tissues causing toxicity.

Q45. B

Visual disturbances are classic signs of Digitalis toxicity.

Q46. B

Narrow therapeutic index necessitates careful monitoring.

Q47. B

Plant taxonomy focuses on identification and classification.

Q48. C

Cryptogams reproduce by spores.

Q49. C

Monocotyledons have parallel venation.

Q50. C

Species is the lowest and most specific taxonomic rank.

End of Model Answers & Detailed Explanations